Cortical Resolution Enhancement in P300 Using eLORETA Technique for BCI Construction

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***Abstract*— Individuals who suffer from severe loss of motor function but retain preserved consciousness often lose the ability to communicate with the external environment. Due to the extent of their motor impairment, these individuals are frequently unable to perform even minimal voluntary movements, further limiting their capacity to interact with others. Brain-computer interfaces (BCIs) have emerged as a promising alternative to restore this communication. However, building robust and accurate BCIs still faces several challenges, among them the low cortical resolution of electroencephalographic (EEG) signals, which, although non-invasive, tend to reflect the activation of large cortical areas—even in response to localized stimuli. This study aims to apply the Low Resolution Electromagnetic Tomography (eLORETA) technique to reconstruct the neural sources underlying EEG signals, with the goal of reducing the region of interest and improving the spatial accuracy of the estimated brain activity. Additionally, it seeks to demonstrate that the P300 event-related potential, combined with the use of high-frequency tactile stimuli, represents an efficient strategy for obtaining robust brain responses in applied contexts. Through the application of this technique to experimental data with tactile stimuli, it was possible to demonstrate that eLORETA significantly improves the localization of active brain regions, proving to be a viable tool for the development of more accurate and robust BCIs.**

Keywords — Brain-computer interface, Electroencephalogram, P300, eLORETA, Cortical Resolution

I. Introduction

Motor function loss caused by accidents or diseases can significantly limit communication for those who were affected, as they may remain fully conscious but become unable to express themselves. This condition is known as locked-in syndrome [1]. Brain-computer interfaces (BCIs) offer an alternative that enables these individuals to communicate with the external environment [2]. Through electroencephalography (EEG), this can be achieved in a non-invasive manner and with high transmission speed, making it EEG a popular method for BCI development [3].

Evoked potentials can be recorded through electrodes placed on the scalp and are capable of capturing responses to specific stimuli applied to an individual. Unlike spontaneous EEG activity, which reflects ongoing brain dynamics, evoked potentials are time-locked responses specifically triggered by external sensory, cognitive, or motor events. These stimuli can vary in nature, for exemple visual, auditory, or tactile. For BCI construction, P300 waves are commonly chosen. These waves are characterized by a positive shift in cortical signal detected approximately 300 milliseconds after a target stimulus, and are selected due to their robustness and reliability [3] [4].

In the development of BCIs, tactile stimuli have gained increasing relevance due to their practical advantages over visual and auditory modalities. Unlike visual stimuli, which can cause fatigue or become unusable in cases of partial or total vision loss, tactile stimuli allow users to keep their eyes free [5]. Similarly, they offer an advantage over auditory stimuli, which are susceptible to interference from ambient noise, potentially compromising performance [5].

Several previous studies have demonstrated the effectiveness of tactile-based approaches. For example, T. Kaufmann, [6] showed that P300 signals could be effectively obtained via tactile stimulation in patients with classical locked-in syndrome. Additionally, C. Chu [7] demonstrated that such methods are stable and highly effective.

The work presented by [8] further supports this, showing through statistical analyses, including analysis of variance (ANOVA), that EEG data collected using tactile stimuli combined with P300 windowing were consistent and viable for EEG signal extraction. Although this finding is already supported in the literature, the confirmation through practical statistical studies strengthens the case for building reliable and robust BCIs.

Nonetheless, EEG signals still face limitations when it comes to brain source localization, often activating brain regions much larger than desired. This poses a challenge for building effective BCIs. To address this, Electromagnetic Source Imaging (ESI) techniques, such as eLORETA (Low Resolution Electromagnetic Tomography), improve the spatial precision of source localization, providing a more detailed view of brain activity [9], which could lead to the development of BCIs with much higher accuracy.

Therefore, the aim of this study is to demonstrate that

applying the eLORETA technique to EEG data collected through tactile stimuli, specifically in the P300 time window, can provide high cortical resolution for brain source reconstruction, reducing signal dimensionality and thus enabling the construction of BCIs with elevated precision.

II. Methodology

A. Data Acquisition

Data were obtained with the participation of 10 healthy volunteers, with no history of mental or nervous system disorders. The EEG signal was sampled at a frequency of 1024 Hz, using 60 electrodes positioned according to the International 10-10 System, and the recordings were made using the BE Plus LTM74 EEG device in conjunction with the MCSCAP 10-10 cap. The experiment was approved by the Ethics Committee of the Universidade Federal de Uberlândia (UFU), under protocol number: 17164719.4.0000.5152.

B. Experimental Procedure

Data collection was conducted by placing two electrodes on the left and right index fingers of each volunteer, with the anode positioned on the distal phalanx and the cathode on the medial phalanx. These electrodes were made of silicone carbon, each measuring 1.5 cm in width and length.

The electrical stimuli were generated by the Neuropack S1 device (Nihon Kohden Co., Tokyo, Japan) and consisted of square waves with frequencies of 10, 25, 50, 100, and 125 Hz, a pulse width of 1 ms, and an amplitude defined based on protocols to determine sensory and pain thresholds.

To assess the sensory and pain thresholds of each volunteer, the frequency was fixed at 10 Hz for the sensory threshold and 125 Hz for the pain threshold. Since the pulse width was also constant, only the current amplitude was varied until the volunteer reported a light sensation on the finger surface (sensory threshold), and then discomfort (pain threshold). This protocol was repeated three times for each finger, alternating the stimuli. Based on the results, the stimulation current amplitude was set at 80% of the average pain threshold for each finger.

The protocol consisted of 25 blocks, five for each frequency value, presented to the volunteer in random order. Each block was composed of 10 trials, with each trial consisting of 1 second of stimulation followed by 4 seconds of rest. Participants remained with their eyes closed and avoided any muscle movement throughout the experiment, including eye movements and blinking, while mentally counting the number of stimuli presented.

C. Data Preprocessing

Preprocessing is an essential step, as raw electroencephalogram (EEG) data typically have a low signal-to-noise ratio, which can hinder source reconstruction and, consequently, the construction of reliable brain-computer interfaces (BCIs).

Data were filtered using a linear trend filter and a Butterworth band-pass filter between 0.5 and 20 Hz, with 4th-order high-pass and 6th-order low-pass filters. Subsequently, epochs were extracted from -200 to 1000 milliseconds, with baseline correction using the pre-stimulus signal.

An Independent Component Analysis (ICA) filter was then applied, with automatic component exclusion using the SASICA algorithm [10], employing the ADJUST method to detect autocorrelation, focal components, and muscle activity [11]. Finally, epochs with amplitudes exceeding 50 µV were removed, and P300 features were extracted using a time window from 250 to 400 ms.

D. Data Processing

In the processing stage, the average of the epochs for each of the five frequencies (10, 25, 50, 100, and 125 Hz) was computed across the 10 volunteers. Using the time window between 250 and 400 ms for the P300, the average response over time was determined for all 60 electrodes, allowing the generation of a topographic map of cortical activity.

The next step involved identifying the exact moment (in milliseconds) when the P300 activity peaked. A function was developed to display the activity of electrodes of interest, such as Cz, over the specified time interval. This allowed the exact peak to be visualized and defined as the latency value for the subsequent analyses.

Next, as an initial model, a dipole model was computed, particularly for the 125 Hz frequency, which as observed in the statistical analysis article [8], offered the best visualization of activity. This model was applied to all 10 volunteers and both fingers to assess whether the signal origin at the latency time corresponded to a physiologically coherent location.

Subsequently, the Low Resolution Electromagnetic Tomography (eLORETA) technique was applied to address the EEG inverse problem using the following regularization function:

$$Q(S) = S^T \times W \times S \tag{1}$$

In this equation, S represents the matrix of estimated brain source signals, and W is a symmetric weight matrix computed based on the head model and the leadfield, which describes how each source influences the EEG electrodes [9]. The goal of eLORETA is to find the signals S that best explain the EEG data while ensuring a spatially coherent distribution of brain activity.

For the processing, the EEGLAB and FieldTrip toolboxes were used. Trials from the 10 volunteers were combined into a single structure with 60 EEG channels, time series, and 500 trials per finger (50 from each volunteer), increasing the expressiveness of dipole values in the regions of interest.

A 3D sourcemodel with 5 mm resolution was then created, covering the entire brain volume. The resulting grid contained 12,213 dipoles inside the brain, distributed according to a standardized anatomical model. The electrodes were projected onto the scalp surface, and their locations were

used to compute the leadfield, which describes the influence of each dipole on each EEG channel.

The leadfield matrix was built considering the conductivity of three head compartments: scalp, skull, and brain, as well as the 3D orientation of dipoles. The EEG data were then processed using the eLORETA method to estimate the distribution of cortical current density. The calculation was regularized using the noise covariance matrix, and the pseudo-inverse was applied to address numerical conditioning issues. In the end, estimates of internal brain activity sources over time were obtained.

E. Data Post-processing

In the post-processing stage, a topographic map was generated showing the distribution of activity among the electrodes, based on the average P300 interval across volunteers. One map was created for each frequency (10, 25, 50, 100, and 125 Hz) and for each finger, resulting in 10 visual maps. A 3D brain model was also created, showing all 12,213 distributed dipoles and illustrating the application of the eLORETA technique.

Finally, after processing with eLORETA, an MRI image of a brain (standard FieldTrip model) was generated, highlighting the region of highest activity for the volunteer data.

III. Results

The results presented in this section stem from the processing stage described previously. They are organized into the following subsections, which address, respectively: the latency value found for the P300 interval; the brain map showing the positioning of the 12,213 dipoles estimated within the brain; and finally, a comparison between the topoplot of the averaged EEG data for the 60 electrodes and the source reconstruction resulting from the eLORETA method.

A. Optimal latency in the P300 interval

Initially, the average stimulus response curve was generated from the combined data of the 10 volunteers. The dataset totaled 500 *trials*, over which the average was computed to obtain the representative curve.

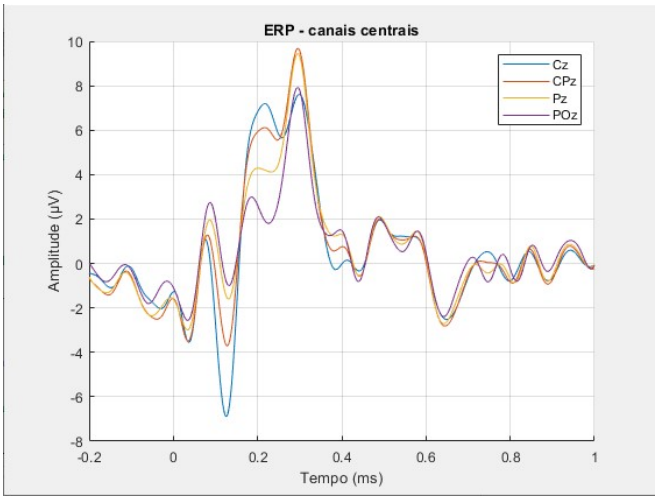


Fig. 1. Activation curves for the combined data of the volunteers

Figure 1 shows the X-axis representing time, ranging from -0.2 to 1 second, corresponding to the duration of a single *trial*. The Y-axis shows the amplitude values in microvolts (µV), ranging from -8 to 10 µV. The channels analyzed in the activation curve were: Cz, CPz, Pz, and POz, regions commonly associated with responses to tactile stimuli applied to the right side of the body.

TABLE 1. Maximum peak values for amplitude and latency at electrodes of interest

Electrode	Latency(ms)	Amplitude(µV)
Cz	297	7.50
CPz	297	9.67
Pz	297	9.40
POz	301	7.92

B. Dipole positioning for use in eLORETA

A three-dimensional *sourcemodel* was generated to cover the brain volume. The constructed grid contains 12,213 dipoles located within the brain, whose distribution can be visualized in Figure 2.

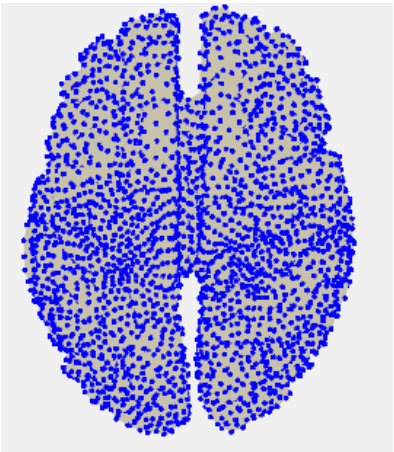


Fig. 2. Distribution of dipoles within the brain volume

The image represents the spatial distribution of the dipoles, with each blue dot corresponding to a position where brain electrical activity was estimated. The visualization is from a top-down angle (view from above the brain). These points were automatically generated based on a standardized

anatomical model using the functionalities provided by the FieldTrip toolbox.

C. *Comparison of EEG results before and after applying eLORETA*

Finally, a comparison was made between the standard topographic map (topoplot) of the average EEG signals from the 10 volunteers, calculated for the 60 electrodes, and the result of the brain source reconstruction obtained after applying the eLORETA method.

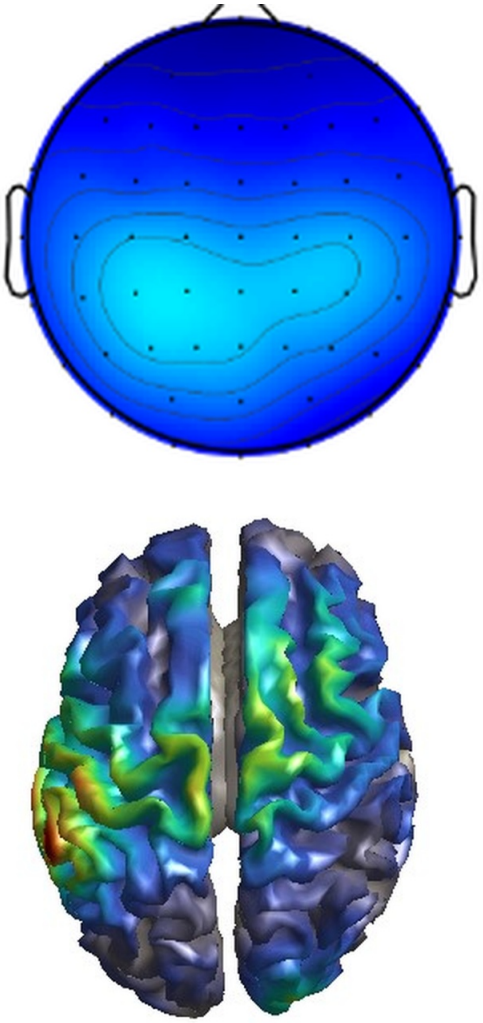


Fig. 3. Top: topoplot of the volunteer average for the 10 Hz frequency; Bottom: source reconstruction using the eLORETA method for the same frequency

In the upper part of Figure 3, the topographic map represents the levels of electrical activity on the scalp surface: blue regions indicate lower activity, while warmer colors (dark red) indicate higher activity. The lower image shows the three-dimensional brain model, viewed from a top-down angle, using the same color scale. This representation highlights the brain regions with the highest response intensity for the analyzed frequency, as estimated by the eLORETA technique.

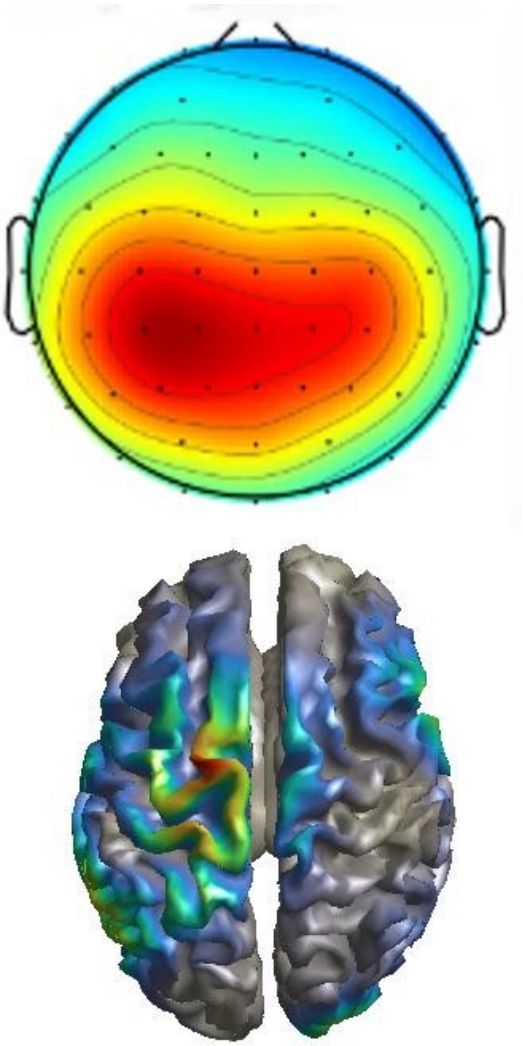


Fig. 4. Top: topoplot of the volunteer average for the 125 Hz frequency; Bottom: source reconstruction using the eLORETA method for the same frequency

IV. Discussion

The findings of this study consistently confirm the feasibility of using the P300 evoked potential as a neurophysiological marker in tactile stimulation paradigms. The analysis of the signals revealed that the maximum peak of the response to the stimulus applied to the participants right index finger occurred approximately 297 milliseconds after stimulation, a value that aligns perfectly with the characteristic temporal window of the P300. This result is in agreement with previous evidence in the literature [3] [4], reinforcing the robustness, inter-subject reliability, and applicability of this potential in Brain-Computer Interface (BCI) systems, especially those requiring fast, consistent, and easily detectable responses.

It was found that high-frequency stimuli, such as 125 Hz, resulted in more localized and intense brain activation patterns, thereby facilitating the identification of cortical regions of interest. In contrast, low-frequency stimuli, such as 10 Hz, produced diffuse and widely spread activation across the cortex, making it difficult to precisely delimit the sources generating the response. These results demonstrate that stimulus frequency not only influences the magnitude

of the brain response but also directly impacts the ability to reconstruct and interpret the involved sources, a critical factor for the development of BCIs with a high degree of spatial precision. The work by Kampe [12], despite using a different stimulation protocol, supports the understanding that higher frequencies elicit a more precise activation area. Furthermore, the study by Alouit [13], which used a protocol similar to that of the present study for stimulation of the right index finger, but applied both electrical and vibrotactile stimuli, found that the type of stimulus itself generated differences in activation precision. However, for vibrotactile stimuli, different frequencies, high and low, did not produce differences in precision, suggesting that, with further research, this could be a viable option for constructing BCIs using this type of stimulus.

Regarding brain source estimation, the eLORETA reconstruction technique stood out by providing a substantial gain in the spatial resolution of the regions of interest, in agreement with the conclusions of E. Schmiele's study [9]. When comparing conventional topographic maps derived from raw EEG with the three-dimensional volumetric reconstruction obtained after applying eLORETA, a significant refinement in cortical activity localization was observed, with the main activation area being much smaller than those previously obtained. This finding may indicate a more specific detection of the stimulated region, such as a single finger rather than the entire hand. This finer discrimination of cortical sources has direct implications for building BCIs that are more selective and responsive, with greater discriminative capability between distinct commands.

The spatial resolution of the model was set at 5 mm, which proved adequate by offering an effective balance between anatomical detail and computational feasibility. Tests with larger resolutions were not efficient in reducing dimensionality, covering larger-than-expected areas, while tests with smaller resolutions, such as 3 mm, increased the number of dipoles to more than 50,000 without improving the identified activity region. This suggests that, even with a 5 mm resolution, the maximum possible reduction was achieved for this dataset using eLORETA, also yielding the lowest computational cost compared to smaller values.

Despite the satisfactory performance of the technique, some methodological limitations must be considered. The anatomical accuracy of the brain resonance model, based on a standardized structure from the FieldTrip library, showed a slight displacement in the estimated source locations. Although the estimated activation occurred in a region compatible with the medial portion of the contralateral somatosensory cortex, as expected for tactile stimuli, a slightly more lateralized activation would be anticipated. This discrepancy may be attributed to the absence of individualized anatomical modeling, as the standard model does not account for subject-specific morphological variations.

Another limiting factor concerns the simplification inherent to the eLORETA model regarding head tissue conductivity. The technique assumes homogeneous conductivity properties within each compartment, the scalp, skull, and brain, which may introduce bias in reconstructing sources near the boundaries of these compartments. Such inaccuracies may be particularly relevant in studies focused on analyzing deep structures or regions adjacent to tissue interfaces with distinct conductivities.

Nevertheless, the data obtained in this work consistently demonstrate the potential of eLORETA to overcome the spatial limitations of conventional EEG, enabling volumetric reconstruction with greater precision and sensitivity. The combined application of adequately parameterized stimuli and high-resolution reconstruction models represents a significant advance in the context of tactile BCIs, making it possible to implement more efficient, adaptable systems compatible with users who have severe motor or sensory impairments.

V. Conclusion

Based on the study conducted and the results obtained, it can be concluded that the application of the eLORETA technique is a promising tool for enhancing cortical resolution in EEG data, providing a consistent foundation for the development of more precise and robust brain-computer interfaces (BCIs).

For future work, it is recommended to integrate structural magnetic resonance images from the volunteers themselves, enabling a more accurate mapping of anatomical structures and, consequently, greater spatial accuracy in source estimation. Furthermore, it is worth exploring other forms of tactile stimulation to reduce the need for very high frequencies in order to achieve responses with a high degree of precision.

In summary, the results obtained reinforce the potential of the eLORETA technique as an enhancement to conventional EEG, significantly expanding the usefulness of such data in applied contexts, particularly in the development of sensitive, precise, and individually adapted brain-computer interfaces.

Conflict of interest

The authors declare that they have no conflict of interest.

Acknowledgments

The authors would like to thank the National Council for Scientific and Technological Development (CNPq), the Coordination for the Improvement of Higher Education Personnel (CAPES), and the Research Support Foundation of the State of Minas Gerais (FAPEMIG) for the financial support, as well as the Federal University of Uberlândia (UFU) and the Biomedical Engineering Laboratory (BIOLAB) – Faculty of Electrical Engineering (FEELT) – UFU for providing the physical infrastructure.

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